Metric Spaces: Topology, Completion, and the Price of Infinity

N-20221218

Two spaces may have the same open sets and yet disagree about which sequences are Cauchy. This small surprise marks the boundary between topology and metric geometry. Starting with convergence and open balls, we can locate exactly where completeness and compactness use information that open sets alone cannot provide.[1]

§1.1 What a metric contributes

A metric $d : X \times X \to [0, \infty)$ is symmetric, vanishes precisely on the diagonal, and satisfies

$$d(x, z) \le d(x, y) + d(y, z).$$

Its open balls $B_r(x) = \{y : d(x, y) < r\}$ define a collection of open sets. A sequence converges to x exactly when its terms eventually lie in every ball around x. Continuity of $f : X \to Y$ means preimages of open sets are open, or equivalently in metric spaces, that $x_n \to x$ implies $f(x_n) \to f(x)$.

For a normed vector space, the metric is $d(x, y) = \|x - y\|$. The Euclidean, taxicab and maximum norms on $\mathbb{R}^n$ generate the same open sets because

$$\|v\|_\infty \le \|v\|_2 \le \sqrt{n}\,\|v\|_\infty, \qquad \|v\|_2 \le \|v\|_1 \le \sqrt{n}\,\|v\|_2.$$

These inequalities explicitly allow a ball for one metric to contain a suitably smaller ball for the other. Their exact distances differ, but they agree on all convergent sequences and continuous functions.

§1.2 One topology, two notions of being complete

Let $X = \mathbb{R}$. Besides the ordinary metric $d(x, y) = |x - y|$, define

$$\rho(x, y) = |\arctan x - \arctan y|.$$

Since $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is a strictly increasing homeomorphism, ρ is a metric inducing the usual open sets.

Nevertheless, $x_n = n$ is a ρ-Cauchy sequence: $\arctan n \to \pi/2$, so

$$\rho(n, m) = |\arctan n - \arctan m| \longrightarrow 0 \qquad (n, m \to \infty).$$

It does not converge in X for ρ, since no real number has arctangent $\pi/2$. Therefore $(\mathbb{R}, \rho)$ is incomplete, although $(\mathbb{R}, d)$ is complete. Even boundedness changes: $\rho(x, y) < \pi$ for all x, y.

A topological homeomorphism preserves neighborhood structure, but need not preserve the comparison of two points moving together. Uniformly continuous maps do preserve Cauchy sequences; ordinary continuous maps need not. That is the extra structure carried by the uniformity of a metric.

[1] See MIT 18.S190, *Introduction to Metric Spaces*, Lectures 2–5: https://ocw.mit.edu/courses/18-s190-introduction-to-metric-spaces-january-iap-2023/pages/lecture-notes/.

§1.3 Completion adds the missing limits in a canonical way

For any metric space (X, d), consider the set $\mathcal{C}(X)$ of Cauchy sequences and identify

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad d(x_n, y_n) \to 0.$$

The distance

$$\widehat{d}([(x_n)], [(y_n)]) = \lim_{n\to\infty} d(x_n, y_n)$$

is well defined: the absolute difference between successive cross-distances is bounded by $d(x_n, x_m) + d(y_n, y_m)$, so those distances form a Cauchy sequence in $\mathbb{R}$. Equivalent representatives give the same limit by the triangle inequality.

Constant sequences embed X isometrically as a dense subset of its completion $\widehat{X}$. Every Cauchy sequence in $\widehat{X}$ converges; one proves this by approximating its successive terms with points of X to within 2^{-n}, and taking the corresponding diagonal sequence. Thus $\widehat{X}$ is complete.

The metric ρ above makes this construction almost visible: $(\mathbb{R}, \rho)$ is isometric to $(-\pi/2, \pi/2)$ with ordinary distance, so its completion is the closed interval $[-\pi/2, \pi/2]$. Two ideal endpoints are added at what had been $+\infty$ and $-\infty$.

§1.4 Compactness requires finite resolution at every scale

A metric space is totally bounded if for each $\varepsilon > 0$ it can be covered by finitely many balls of radius ε. Mere boundedness only provides one large ball. An infinite set with the discrete metric is bounded but not totally bounded: every ball of radius less than 1 contains a single point.

Theorem 1.4.1

A metric space is compact if and only if it is complete and totally bounded.

The constructive direction is particularly instructive. Suppose X is totally bounded. Cover X by finitely many balls of radius $1/2$; infinitely many terms of any sequence (x_n) lie in one of them. From those indices, choose an infinite subsequence lying in one ball of radius $1/4$. Repeat with radii 2^{-k} and take a diagonal subsequence. Its tail lies in a ball of radius 2^{-k} for every k, so it is Cauchy. If X is complete it converges.

Thus complete plus totally bounded implies sequential compactness, which is equivalent to open-cover compactness for metric spaces. In the other direction, compactness forces finite ball covers immediately, and compactness also implies completeness by extracting a convergent subsequence from each Cauchy sequence.

§1.5 Closed and Bounded in Infinite Dimensions

In $\mathbb{R}^n$, closed and bounded sets are compact because bounded sets are totally bounded and closed subsets of a complete space are complete. The missing implication in infinite dimensions is total boundedness.

In the Hilbert space ℓ^2, let e_k be the sequence with a single 1 in position k. Every e_k belongs to the closed unit ball, but

$$\|e_j - e_k\|_2 = \sqrt{2} \qquad (j \neq k).$$

No ball of radius $1/2$ can contain two of these points, so the unit ball cannot be covered by finitely many such balls. It is closed and bounded, yet not compact.

This obstruction motivates weak topologies in functional analysis. The closed unit ball of a Hilbert space need not be compact in the norm topology, while it is compact in the weak topology by a more sophisticated theorem (Banach–Alaoglu, together with reflexivity). Changing the topology deliberately changes which kinds of compactness are available.

§1.6 Completeness also makes dense properties coexist

The Baire category theorem states that a complete metric space is a Baire space: a countable intersection of dense open subsets is dense. To see the mechanism, take dense open sets $U_1, U_2, \dots$ and any nonempty open ball B_0. Since U_1 is dense and open, one can choose a closed ball B_1 of positive radius, contained in $B_0 \cap U_1$. Continue recursively with

$$B_n \subseteq B_{n-1} \cap U_n, \qquad \operatorname{radius}(B_n) < 2^{-n},$$

and nonempty interior for each ball. Their centers form a Cauchy sequence. Completeness gives a limit lying in every closed B_n, hence in $B_0 \cap \bigcap_n U_n$.

The proof shows exactly what completeness buys: a nested sequence of increasingly accurate approximations produces an actual point. For Baire category, this point simultaneously satisfies countably many dense conditions; for metric completion, it realizes a Cauchy sequence; for compactness, finite approximations at every scale provide a convergent subsequence. The three theorems share a concrete construction rather than just a similar-sounding notion of convergence.